

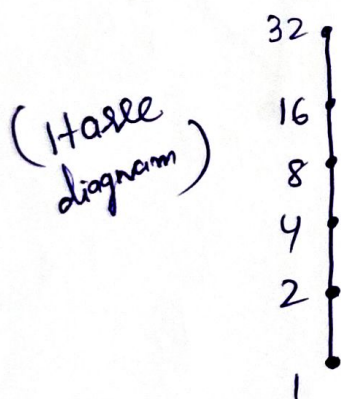
Bounded Lattice, complemented Lattice and distributive Lattice.

Bounded Lattice

A lattice is said to be bounded lattice if it has both least and greatest element.

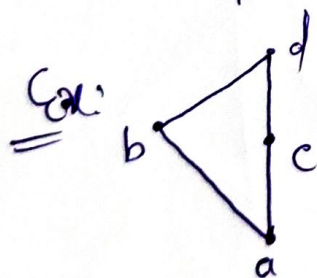
- Every finite lattice has a least element and a greatest element.
- Every finite lattice is bounded.

Ex: • Divide relation on $\{1, 2, 4, 8, 16, 32\}$ is a bounded Lattice.

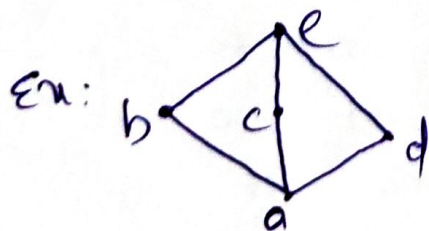


least element is 1
greatest element is 32

So, the lattice $(\{1, 2, 4, 8, 16, 32\}, 1)$ is a bounded lattice.



It is a bounded lattice as it has both least and greatest element.
least element is a
greatest element is d.



$\mathbb{Z}^+$ is a bounded lattice.
greatest element is e
least element is a.

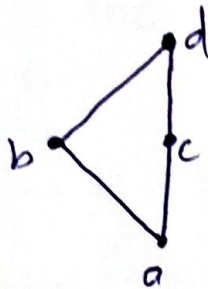
Ex The poset $(\mathbb{Z}^+, 1)$ is a lattice but not a bounded lattice as greatest element does not exist.

complemented Lattice

A complemented lattice is a bounded lattice where every element in the lattice has at least one complement.

- a is called the complement of b if both the following conditions are satisfied:
 - (i) $\text{glb}(a, b) = \text{least element}$
 - (ii) $\text{lub}(a, b) = \text{greatest element}$.
- least and greatest elements are complement to each other.

Ex: Check whether the given lattice is complemented or not.



It is a bounded lattice with least element (a) and greatest element (d).

complement of a

$$\begin{aligned}\text{lub}(a, d) &= d = \text{greatest} \\ \text{glb}(a, d) &= a = \text{least}.\end{aligned}$$

$\therefore d$ is the complement of a and a is the complement of d .

complement of b

$$\text{glb}(b, c) = a, \text{lub}(b, c) = d$$

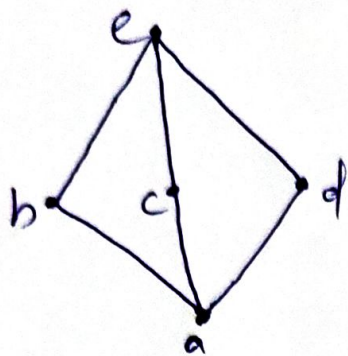
$\therefore c$ is the complement of b and b is the complement of c .

one can verify that a is not a complement of b since $\text{glb}(a, b) = b \neq a$.

one can verify that d is not a complement of b since $\text{lub}(d, b) = b \neq d$.

So the given lattice is a complemented lattice as every elt has at least one complement.

Ex.



Is the given lattice is complemented?

It is a bounded lattice with least (a) and greatest (e).

a and e are complement to each other.

complement of b

$$\begin{aligned} \text{glb}(b, c) &= a \\ \text{lub}(b, c) &= e \end{aligned}$$

$\Rightarrow$ c is the complement of b.

Also b is the complement of c.

$$\begin{aligned} \text{glb}(b, d) &= a \\ \text{lub}(b, d) &= e \end{aligned}$$

$\Rightarrow$ d is the complement of b.
Also b is the complement of d.

[one can verify e is not a complement of b and a is not a complement of b].

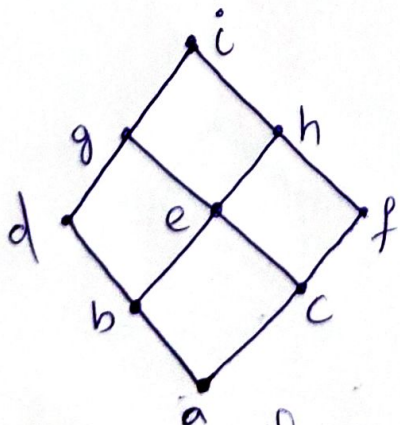
$\therefore$ b has two complements c and d.

Similarly, c has two complements b and d.

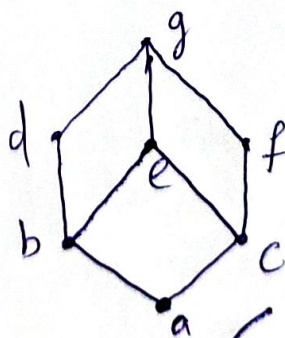
d has two complements b and c.

Since every element has atleast one complement,
given lattice is a complemented lattice.

Ex: The following lattices are bounded but not complemented.



(complement of e does not exist)



(complement of e does not exist)

Distributive Lattice

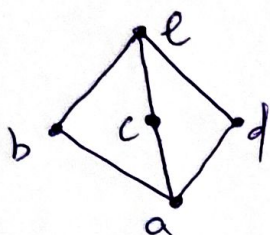
A lattice L is said to be distributive lattice if $\forall a, b, c \in L$, the following properties hold:

$$(i) a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$$

$$(ii) a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$$

Notation: $p \vee q = \text{lub}(p, q)$, $p \wedge q = \text{glb}(p, q)$.

Q. check the given lattice is distributive or not.



consider $b, c, d \in L$

$$\text{Now } b \vee (c \wedge d) = b \vee a = a$$

$$\text{and } (b \vee c) \wedge (b \vee d) = e \wedge e = e$$

$$\text{So } b \vee (c \wedge d) \neq (b \vee c) \wedge (b \vee d)$$

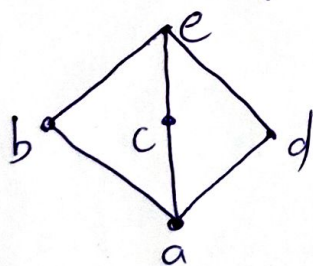
Thus given lattice is not a distributive lattice.

Note: A given lattice is distributive if both the distributive properties must be satisfied for any three elements in the lattice.

Alternative

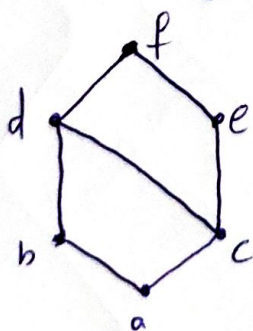
Def: A lattice L is said to be distributive lattice if every element in L has at most one complement.

Ex:



It is not a distributive lattice as b has two complements c and d .

Ex:



- a is the complement of f and f is the complement of a .
- b is the complement of e and e is the complement of b .
- complement of c does not exist and complement of d does not exist.

Every element has at most one complement, so it is distributive.

• It is bounded, not complemented but it is distributive.